

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

Coordinated cortical activity need not describe a single experienced situation. We develop a conditional field account of coherence, compatibility and self-representation. Its demonstrated results establish what coordination leaves undetermined: increasing coupling, decreasing diffusion and a matched common drive can produce identical stationary phase distributions; equal phases permit incompatible local states; and a plasticity control preserves coherence while losing alignment with a specified environmental template. The conditional construction glues compatible spatial measures into one state and makes that state the unique fixed point of a contracting reconstruction map. Lean 4 checks the composition under eight connecting hypotheses and additional structures, with compatibility and accuracy at the glued state supplied as premises. We propose the glued state as a formal correlate of conscious unity and its accurate internal reconstruction as a correlate of a minimal self. The physical hypothesis is that changing extracellular geometry increases field-mediated coupling during awakening. A noisy mean-field phase model gives a square-root stationary onset, but finite ramps delay the response and the distributional

equivalence requires measurements that distinguish recovery mechanisms. Testing the account further requires identifying neural content and self-encoding independently, then determining whether coordination maintains their agreement and whether these relations matter for unified experience.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Yet even successful coordination leaves a question: what makes the coordinated activity a description of one experienced situation, with a perspective on that situation?

We propose an account with three requirements. *Coherence* concerns whether coupling sustains collective activity. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting global state supports an accurate model of itself. The latter two express commitments about consciousness whose motivation and possible dissociations we develop below.

Consider the experience of reaching for a cup. A description of the cup's location and a description of the hand's relation to it can share information about where the cup is while contributing different contents to the episode.

19 Synchronous activity in the two processes would leave any disagreement
20 about that shared quantity unresolved. The formal counterexample isolates
21 this problem using spatial mass profiles at a common phase that disagree
22 on an overlap. We return to the cup example to ask what agreement would
23 mean for perceptual content, and what an internal representation would have
24 to add to place the perceiver in that situation. It is an illustration of the
25 modelling task, not a neural model implemented here.

26 Awakening gives the physical requirement a concrete setting. We hypoth-
27 esize that changing extracellular geometry slowly increases effective coupling
28 through a synchronization threshold. Under stationary branch tracking, a
29 noisy phase model fixes a square-root onset of coherence. The experiment
30 faces two distinct obstacles. Finite ramp speeds delay the response, so a
31 fitted onset exponent depends on the drive. Moreover, successive stationary
32 phase distributions can be identical under increasing coupling, decreasing
33 diffusion or a suitably matched common drive. Recovery must therefore be
34 measured in a way that identifies dynamics and distinguishes causes, rather
35 than inferred from the shape of a coherence trace alone.

36 A plasticity control tests another route to representation. Its squared-
37 drift objective rewards equalizing drift, which in the chosen frequency-cluster
38 model favors connections across the clusters that define the diagnostic tem-
39 plate. Coherence survives while alignment with that template declines. To-
40 gether with the overlap counterexample, this result establishes a limit on
41 what the coordination model explains. Compatible local content needs its

own mechanism and observable.

The contribution has three levels. The separation results establish limits on what coordination explains. The conditional construction gives a unique global state from specified compatible spatial measures and makes it the unique fixed point of a contracting map assumed to reconstruct that state. Lean 4 checks the composition, and toy witnesses establish joint satisfiability of its assumptions. The biological interpretation proposes that the glued state and its internal reconstruction capture conscious unity and a minimal self. This last step requires identifying meaningful neural descriptions and explaining how coordination maintains their agreement. Figure 1 locates the proposed connections, and Section 6 states the physical and thermodynamic assumptions.

2 Coherence: the coupling model

2.1 Dynamics and the stationary threshold

The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles dimensionless. The uniform mean-field convention is $K_{ij} = K/N$. Frequency heterogeneity and phase diffusion are

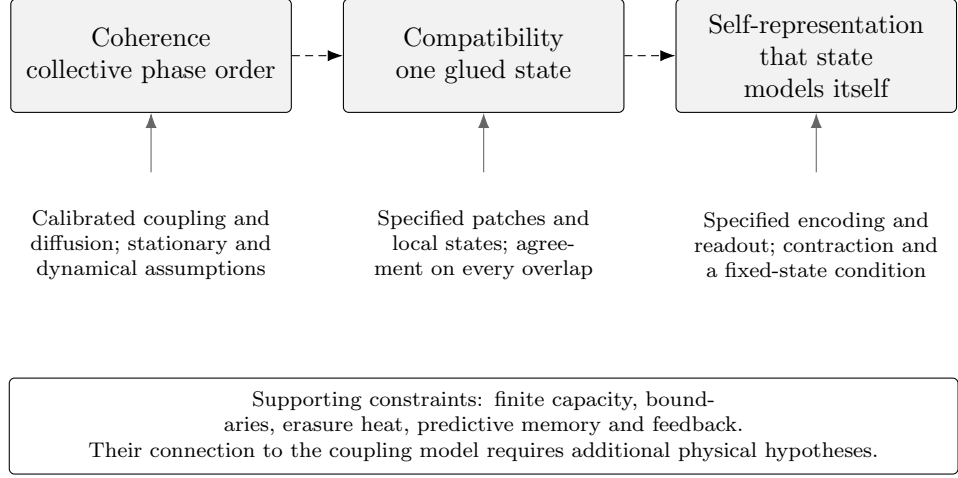


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

60 separate parameters.

61 For identical frequencies in their rotating frame, the positive normalized
62 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

63 where I_n are modified Bessel functions and ψ is the collective phase. Lean
64 derives this form rather than positing it: a periodic gradient drift carries no
65 stationary current, and the resulting first-order equation leaves the Gibbs
66 weight as the only positive normalized solution. Lean also verifies normal-

67 ization, the circular order parameter and self-consistency for this density.
68 At $K \leq K_c = 2D$, the only nonnegative self-consistent order parameter is
69 zero. At $K > K_c$, exactly one positive solution coexists with zero; it emerges
70 continuously and increases strictly with K . Near threshold,

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

71 These results characterize the stationary branch (Fig. 2). Linearizing the
72 same equation about each branch gives the threshold a dynamical reading.
73 About the uniform density the coupling reaches only the first harmonic,
74 whose rate is $K/2 - D$, while the n -th decays at $-Dn^2$, so a growing mode
75 exists exactly above K_c . About the coherent density the quadratic form is
76 strictly dissipative transverse to the rotation orbit, and the rate along that
77 orbit is zero, so linear stability holds modulo the collective phase and not on
78 the nose. These are linear estimates for the limiting equation; they construct
79 no solution of it. Connecting it to finite trajectories requires a dynamical
80 mean-field limit and dynamical selection, neither proved here. Propagation of
81 chaos concerns the first of these steps (Sznitman, 1991); exchange symmetry
82 alone does not establish independence, especially when realizations have an
83 unpinned collective phase.

84 A spatial version replaces the finite sum by an integral against a kernel
85 $K(x, y)$. The substrate measure has total mass one, making the mean-field
86 strength a kernel average. Applying the scalar threshold to a heterogeneous

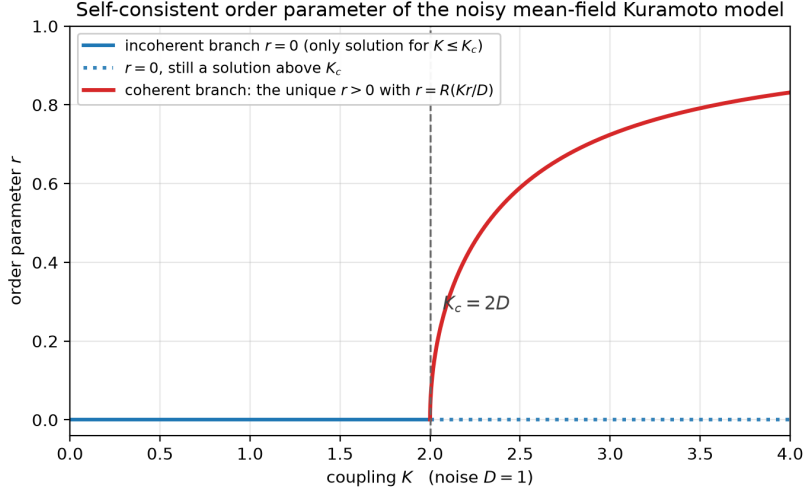


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

87 or signed kernel requires a justified reduction to that mean-field description.
88 Whether such a reduction preserves the organization it describes is itself mea-
89 surable. Calcium imaging across wakefulness, sleep and anesthesia reports
90 higher network modularity under unconsciousness at single-cell resolution,
91 with spatially intermixed modules, and no consistent modularity difference
92 at mesoscale, with spatially localized ones (Kiyooka et al., 2026). A reduction
93 of a cortical kernel has to name the scale at which it claims to hold.

2.2 Physical identification and calibration

The cortical hypothesis identifies part of the effective coupling with endogenous extracellular electromagnetic interactions. Fields can modulate spike timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models have been proposed for coordinating distributed neural information (Pinotsis & Miller, 2022; McFadden, 2020). The quantitative question is whether that interaction supplies sufficient coupling relative to diffusion. Changes in phase organization under propofol (Bastos et al., 2021) motivate studying this regime, but do not measure the coupling itself.

The physical field predicate requires a jointly continuous kernel, a normalized substrate, a mean-field strength equal to the spatial average, and no single site providing more than half that strength. Regular mesh refinement establishes convergence of a scalar coupling-energy quadrature; constructing the kernel and relating it to trajectories are separate modelling tasks. The formal witnesses supply toy instances of the predicate.

Calibration is essential because a spike-time shift is not a coupling rate. An estimate proportional to $N\Delta tf$, using population size, timing shift and oscillation frequency, is dimensionless. A conversion of the form

$$K_{\text{eff}} = \gamma N s E f \tag{4}$$

requires a rate γ , where s is the timing shift per unit field and E the field amplitude. This rate specifies how induced timing shifts persist and accumu-

late into phase drift. Supplemental Material explains why the other factors cannot determine it. The biological threshold test requires independent estimates of the interaction kernel, its aggregation, and phase diffusion in the same cortical regime.

3 Testing coherence through recovery after awakening

The proposed control variable is extracellular geometry. Sleep-associated changes in extracellular space and clearance (Xie et al., 2013) motivate a coupling parameter that evolves more slowly than phase dynamics. The extracellular volume fraction is approximately $\alpha = 0.2$ at diffusion tortuosity $\lambda \approx 1.6$ (Syková & Nicholson, 2008); the cited sleep study reports a rise in α from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz cortical tissue conducts approximately resistively at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). In an effective-medium description at fixed transmembrane current density, a larger extracellular volume fraction raises conductivity and reduces field amplitude.

This gives the coupling-gated hypothesis a directional prediction: contraction of extracellular space on awakening raises E and, with the other factors in Eq. (4) held fixed, raises effective coupling. The proposed geometry-coupling relation and its role in human recovery both require measurement. The effective-medium argument supplies a sign under stated conditions; cal-

135 ibration supplies the magnitude and the mapping to the scalar model.

136 If $K(t)$ rises through K_c and the phase distribution tracks the coherent
137 stationary branch, Eq. (3) gives

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0, \quad \dot{K}(t_0) > 0. \quad (5)$$

138 The crossing time satisfies $K(t_0) = K_c$. It is a prediction of latency only when
139 coupling is measured independently of coherence. The subsequent shape
140 and saturation depend on the later coupling trajectory. Critical slowing
141 down restricts branch tracking near threshold, while finite populations have
142 fluctuating order below it. A measurable onset therefore needs an explicit
143 criterion above that fluctuation floor.

144 **3.1 What a recovery distribution can identify**

145 Equation (2) gives a concentration–coherence relation with no fitted shape
146 parameter. Testing it requires separate estimators: r from circular resultant
147 length and a from density shape. Estimating a by inverting R from the
148 measured r makes agreement automatic. Even a valid agreement tests a
149 statistical description shared by the von Mises family; it does not determine
150 the coupling medium.

151 There is a stronger limitation. Successive equilibrated phase distributions
152 cannot, by themselves, distinguish three different recovery mechanisms. Let
153 $q(t)$ be a prescribed ratio crossing 2, let $r(q)$ denote its stationary branch,

154 and choose a fixed rate $D_0 > 0$. Increasing coupling with $K(t) = D_0 q(t)$
 155 and $D = D_0$ gives the same stationary density as fixed coupling K_0 with de-
 156 creasing diffusion $D(t) = K_0/q(t)$. Both have concentration $a = q(t)r(q(t))$.
 157 Uncoupled oscillators under a common prescribed drift $-h(t) \sin(\theta - \psi)$, with
 158 $h(t) = D_0 q(t)r(q(t))$ and $D = D_0$, have that same density. Their distribu-
 159 tions coincide window by window, even though their physical causes differ.
 160 Supplemental Material supplies the construction and its numerical feasibility
 161 check.

162 The missing observable is the physical clock. Resolved phase increments
 163 contain rate information: in the ideal continuous-path model, quadratic vari-
 164 ation accumulates at rate $2D$, while conditional increment means constrain
 165 drift. This can separate coupling and diffusion under the specified observa-
 166 tion model. Distinguishing endogenous coupling from the matched common
 167 drive additionally requires a measured drive or a controlled perturbation, be-
 168 cause their mean-field drifts can coincide. Identifying the coupling as electro-
 169 magnetic then requires evidence specific to that interaction. This hierarchy
 170 determines the experiment: distribution, dynamics and physical mechanism
 171 are distinct targets.

172 **3.2 Finite ramps change the observable onset**

173 Finite simulations characterize how the stationary prediction can appear in
 174 a recovery measurement. The ramp study uses $N = 2000$, $D = 1$ and 32
 175 replicas at four coupling speeds. Escape is the first of three consecutive

176 saved samples at $r \geq 0.2$ after threshold. The supplement gives parameters,
177 estimators and further controls.

178 Faster ramps delay escape (Fig. 3). Only 5/32 replicas escape within the
179 fastest window. The three slower ramps have mean coupling delays 0.1947,
180 0.0684 and 0.0253. Their fitted onset exponents run from 0.898 to 0.417,
181 compared with 0.443 to 0.448 from the same estimator on the exact stationary
182 branch at the same coupling grids. At the fastest speed the ensemble mean
183 never reaches the declared fit window; the returned value 0.964 is a local
184 log-slope of a delayed trajectory.

185 This is delayed passage through a noisy bifurcation (Berglund & Gentz,
186 2002), with a direct consequence for the awakening test: the apparent expo-
187 nent depends on the ramp and on the fitting window. A deviation from the
188 square-root law constrains the adiabatic approximation only where branch
189 tracking is independently justified. The sweep does not bracket the specified
190 deviation-based critical ramp speed.

191 Additional controls test the approximations used by the physical model.
192 Time-step refinement attributes most of the sampled discrepancy from the
193 stationary order to integration bias. Population-size diagnostics show why
194 a shared random collective orientation can preserve unconditional pair cor-
195 relation in the coherent regime. Spatial and signed-network controls exhibit
196 coherence at specified total coupling, with the refined signed-network cross-
197 ing bracket $K_{\text{eff}}/D \in [1.958, 2.009]$ still containing 2. These controls delimit
198 finite regimes and numerical resolution; the supplement reports their full

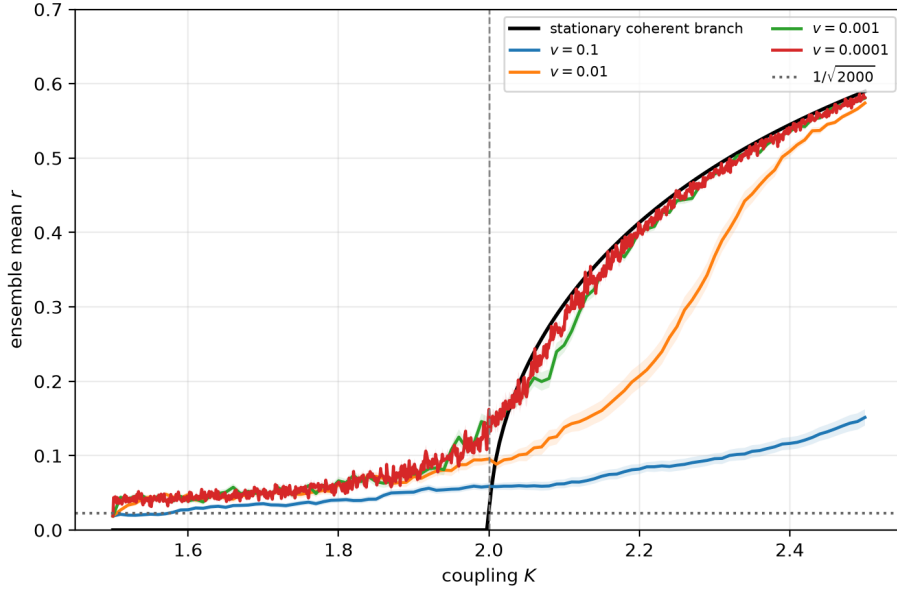


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

199 evidence. Calibrating the cortical coupling remains the experimental task.

200 3.3 What the estimators produce on EEG

201 Public propofol-sedation EEG (Bajwa et al., 2025) provides an exploratory
 202 exercise of the concentration and coherence estimators. The implemented
 203 analysis pools channels over 100-ms windows, estimating a spatiotemporal
 204 phase distribution. Bipolar and common-reference montages produce differ-
 205 ent residuals. The bipolar results are compatible with the Bessel relation,
 206 but the concentration range is too narrow to distinguish it from its linear

207 approximation at the observed scatter.

208 An informative recording needs a broader concentration range, estima-
209 tor calibration that accounts for dependence, and an observation model for
210 source mixing, referencing and temporal pooling. The sleep-recovery hypoth-
211 esis remains untested by these sedation recordings. Public single-cell calcium
212 imaging of mouse cortex supplies the site count and the state coverage they
213 lack, recording thousands of neurons per session with their spatial coordi-
214 nates across wakefulness, NREM and REM sleep and isoflurane anesthesia
215 (Oomoto et al., 2026). Its 7.65 Hz frame rate lies below the band in which
216 the phase estimators used here operate, so it addresses sites and states rather
217 than the bandwidth those estimators require.

218 **3.4 A protocol for the awakening experiment**

219 The analysis defines six requirements for the physical test.

- 220 **1. Coupling and diffusion in the same units.** Estimate K and D
221 from resolved phase dynamics in the same cortical regime, separating
222 diffusion, frequency heterogeneity and observation noise. A measured
223 drive or controlled perturbation is additionally needed to distinguish
224 endogenous coupling from the matched common-drive alternative.
- 225 **2. Geometry calibrated to coupling.** Establish the relation between
226 an extracellular-geometry observable and effective coupling, with un-
227 certainty. A diffusion-MRI or astrocytic proxy for volume fraction re-

228 quires its own validation; the conductivity argument supplies only the
229 expected sign.

230 **3. Adequate phase measurements.** For the proposed spatial estima-
231 tor, use at least 100 sites per estimate, r from resultant length, and
232 a from unweighted regression of log density on the cosine of centered
233 phase. This numerical minimum needs null calibration at the actual
234 sample count, concentration range and dependence. Temporal pooling
235 changes the estimand.

236 **4. An independent recovery endpoint.** Measure behavioral recovery
237 separately from the phase trace.

238 **5. A predeclared onset criterion.** Fix a coherence threshold above the
239 finite-size floor and account for finite ramp speed before interpreting
240 an onset exponent.

241 **6. Matched model comparison.** Compare coupling-gated recovery
242 with diffusion change, common drive and alternative slow-recovery
243 models under the same observation and filtering assumptions. Station-
244 ary distributions alone cannot resolve the equivalence in Section 3.1.

245 Failure of the concentration–coherence relation over an informative range
246 would constrain the phase-density model. Failure of the calibrated geometry–
247 coupling association would constrain the proposed physical gate. A com-
248 peting dynamical explanation that better predicts independently measured

249 recovery would count against its role in awakening. These tests address the
250 first requirement of the framework. Even a favorable outcome leaves the
251 question of what the coordinated activity represents.

252 4 Compatibility: from coordinated activity 253 to a global description

254 4.1 Why common timing is insufficient

255 Consider overlapping patches U and V , each carrying a local description.
256 Compatibility requires that restricting either description to $U \cap V$ gives the
257 same object. Agreement concerns the shared domain; descriptions on distinct
258 parts can remain different. For the finite spatial measures used here, equal
259 overlap restrictions give a unique measure on the covered domain. This is
260 the gluing property formalized by a sheaf.

261 In the cup example, suppose a visual description locates the cup on a
262 table and a reaching description relates the hand to the cup. To compare
263 them, a model must express their shared quantity, such as cup location, in
264 a common coordinate system. Restriction would retain that quantity while
265 dropping details peculiar to each description. Agreement on it permits the
266 visual description to contribute surface appearance and the reaching descrip-
267 tion to contribute hand configuration without duplicating the whole situa-
268 tion. Disagreement after coordinate conversion would make the two estimates

269 incompatible at that level of description, however closely timed their activity.

270 The formal counterexample makes the additional requirement visible. Ev-
271 ery patch has the same oscillator phase, yet two local mass profiles assign
272 different densities to a shared site. They cannot be restrictions of a single
273 global measure. Phase order therefore cannot establish compatibility for de-
274 scriptions that are free of the phase, and continuity of a physical field does not
275 repair disagreement in the descriptions associated with it. The separation is
276 exactly this freedom: the counterexample's profiles are chosen independently
277 of the oscillators, so equal phases constrain them not at all. Where the lo-
278 cal description is instead a declared Lipschitz function of the local phase,
279 coherence does bound overlap disagreement, and the bound is stated in the
280 Discussion. The two statements fence each other. Coherence buys agreement
281 for content that reads the phase, and buys nothing for content that does not,
282 so which of the two a neural description is becomes an empirical question
283 rather than a modelling convenience.

284 Agreement to within a tolerance determines less than a glued state, and
285 more than nothing. Suppose the local profiles differ by at most ε in the
286 uniform distance at every shared site. Nonnegative weights that vanish off
287 their own patch and sum to one at each site then select a global profile lying
288 within ε of every patch on that patch, and any two such selections differ by
289 at most ε , with no dependence on how many patches cover a site; at $\varepsilon = 0$
290 the selection is the unique glued state. What the tolerance costs is existence:
291 for every positive ε there are families at that tolerance admitting no global

description at all, so selection returns a choice among states rather than an extension of the data. A readout that is L -Lipschitz in the state varies by at most $L\varepsilon$ across selections, which is the resolution condition an interpretation at scale δ would have to meet. The supplement states these theorems and a three-site witness that attains the bound. Selection supplies no process by which local descriptions come to agree, and no rule that picks the cover or the weights.

For a theory of experience, the attraction of gluing is this contribution of different local contents to one description. The cup example does not yet instantiate the theorem: receptive territories and content domains must be related to the spatial patches, and their descriptions must have the required extension property. In particular, a decoder assigning probability laws to perceptual contents needs a further extension model, because matching overlap marginals need not determine a unique global joint distribution (Abramsky & Brandenburger, 2011). The spatial-measure construction provides a precise case of gluing; a content model must justify its own existence and uniqueness claims.

4.2 Coherent adaptation can lose environmental alignment

An adaptive coupling rule might appear to supply that missing representational structure. The squared-drift control shows why the objective mat-

ters. Symmetric coupling conserves mean drift: pairwise sine terms cancel when summed over nodes. Minimizing the sum of squared drifts at that fixed mean therefore rewards equalizing drift across nodes. In the chosen frequency-cluster model, this favors coupling between unlike-frequency clusters, whereas the diagnostic template rewards coupling within like-frequency clusters. The objective and the scored structure pull in different directions.

For fixed phases, the continuum objective is

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (6)$$

The formalization constructs the bounded drift operator A and its adjoint, obtains $\nabla\sigma = (2/D)A^*(AK + \omega)$, and proves descent for the specified gradient flow. This is a squared-drift functional; identifying thermodynamic entropy production would require probability-current information.

The finite control co-evolves noisy phases and symmetric, nonnegative couplings at fixed total resource. The frequency-cluster template is a diagnostic covariance of shared latent signals, not a training input. Across three seeds, second-half order remains between 0.9167 and 0.9222, while the within-cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 4). The supplement reports random-update, relabelled-gradient and frozen-kernel controls, including comparisons at overlapping deformation and coherence ranges and against partitions blind to frequency.

The objective reduction is transient: averaging over complete update

intervals removes the apparent gain seen immediately after updates. A pre-specified grid of 6 rate–cadence combinations contains no case meeting the declared 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables. This control tests the narrower suggestion that coherent adaptation under this objective acquires the designated environmental structure. It fails for an intelligible reason: the objective rewards a different organization. A thermodynamically specified learning model can include data in its objective (Corberi et al., 2026); explaining agreement about the cup would likewise require a learning rule connected to the relevant content.

4.3 What must be specified and measured

Compatibility enters the construction as a hypothesis about the local states. For deterministic positive coupling, convergence to a potential minimum is proved under an explicit initial basin condition; the proposed bridge E78 connects supercritical noisy coherence to relaxation of the specified cover. Neither step explains how initially inconsistent representations become consistent. That is the mechanism a biological account must add.

A neural test must first identify the domains and objects whose agreement is at issue. Two decoders should estimate a shared content on overlapping domains, with restriction maps specified before inspecting their agreement.

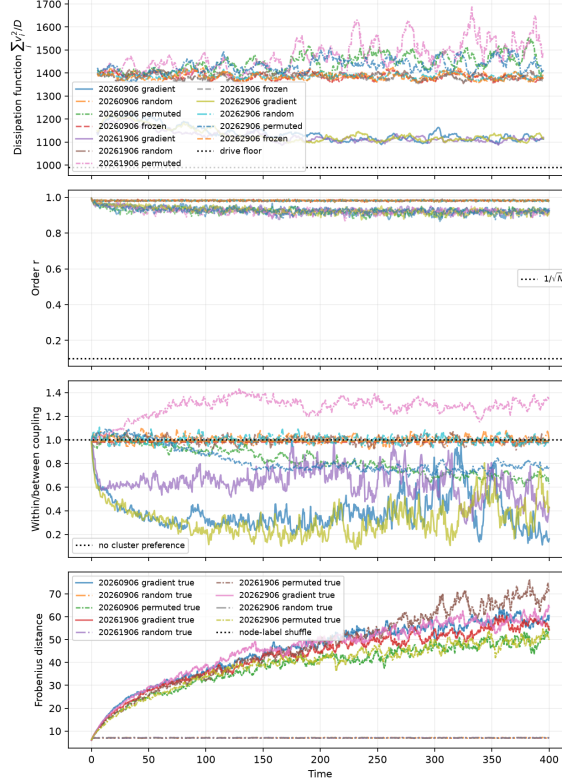


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), relabelled gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

355 Their discrepancy needs a null under decoding noise, and the procedure must
356 distinguish deliberately constructed compatible and incompatible examples.
357 A common prior can make separate decoders agree artificially; region-specific
358 bias can make them disagree artificially. The supplement develops these re-
359 quirements. An estimator built from phase agreement would miss the prop-
360 erty the counterexample separates.

361 Choosing a cover is also part of choosing a candidate subject. Trivial
362 descriptions can agree while carrying little content, and a suitably chosen
363 partition can conceal disagreement. The scientifically useful choice must
364 therefore be constrained independently by what local activity encodes and
365 which shared quantities can be compared. The uniqueness theorem then
366 answers a precise question about that choice: whether the supplied local
367 descriptions determine one global state. It leaves the empirical selection of
368 domains and descriptions to the model.

369 **5 Self-representation: a state that models it-** 370 **self**

371 **5.1 The reflexive construction**

372 Compatibility yields a global state. Reflexivity adds a relation between that
373 state and an encoding within it. Equip global sections with a complete
374 metric. Under the additional restriction-resonance condition, the encoding

375 of a section s is its restriction to a region U , and the map takes the form

$$F(s) = \text{readout}(s|_U). \quad (7)$$

376 The same region and readout are used across the model’s state space. If F
377 contracts and fixes the section s_* supplied by the cover, Banach’s theorem
378 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89 sup-
379 plies the contraction law and $F(s_*) = s_*$ for the boundary’s encoding–readout
380 composite. Identifying that encoding with restriction is a separate resonance
381 condition; E89 does not supply it. The supercritical order parameter pro-
382 vides an inequality used by the specified contraction bound; constructing the
383 encoding and readout remains a separate obligation.

384 The equality ties the reconstruction to the state formed from the local
385 descriptions. In the cup example, the candidate global state would include
386 the represented scene and the perceiver’s relation to it. A local encoding
387 would have to retain enough information for the specified readout to recover
388 that modelled state. An encoding of cup position alone would be insufficient
389 if the readout could not recover the represented hand and viewpoint. The
390 term *predictive* names this reconstruction map; a temporal neural prediction
391 requires a dynamical interpretation.

392 5.2 Why connect these properties to unity and a min- 393 imal self?

394 We identify the section glued from compatible local states with the unity
395 of a conscious episode, and its accurate internal reconstruction with a mini-
396 mal self. Two features of the episode motivate this interpretation: different
397 contents belong to one situation, and that situation is experienced from a
398 perspective.

399 The unity proposal starts from a distinction between having several con-
400 tents and having several experiences. Seeing the cup's surface and feeling
401 the reaching hand contribute different contents to one situation. Compat-
402 ibility expresses their agreement about what they share while preserving
403 those differences. An episode can also contain uncertainty or conflicting
404 judgments. A model might represent the cup's apparent position separately
405 from a judged position and consistently describe the discrepancy. Deciding
406 whether two estimates concern the same quantity or distinct aspects of ex-
407 perience is therefore part of defining the content model. A choice made only
408 to ensure agreement would give the unity proposal no empirical force.

409 The motivation for a minimal self is that the cup is encountered from a
410 location and in relation to a body that can reach it. The proposed internal
411 representation concerns the perceiver's condition in that encounter: where
412 the perceiver is situated, what is being represented and how the body relates
413 to it. Such a representation could be implicit in the organization of activity;

414 it need not describe a person verbally or carry an autobiographical narrative.
 415 Requiring it to reconstruct the same global state links the perspective to the
 416 episode whose contents it organizes.

417 The fixed-point condition requires accuracy about the very state contain-
 418 ing the encoding. Reconstruction alone is insufficient for a perspective: a
 419 recording can be reconstructed without being used by a system to represent
 420 its own condition. The proposed interpretation therefore needs evidence that
 421 the encoding helps the system distinguish changes in itself from changes in
 422 its surroundings. Changing the encoded hand or viewpoint while holding
 423 the external scene fixed should alter the system’s estimates of its relation
 424 to the cup as the readout predicts. This functional criterion goes beyond
 425 $F(s_*) = s_*$; whether the represented perspective is experienced remains a
 426 further question.

427 Accuracy concerns the variables represented by s_* , rather than every mi-
 428 croscopic physical fact. For a fixed map F with Lipschitz factor $q < 1$ and
 429 fixed point s_* , approximate reconstruction has a quantitative bound:

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - q}. \quad (8)$$

430 At the stipulated rate $q = e^{-(K-2D)\tau/2}$, this radius grows as $2\varepsilon/((K-2D)\tau)$
 431 near threshold, for fixed positive ε and τ . The bound weakens while the
 432 exact fixed point remains unique. The supplement gives explicit inequalities
 433 and composes reconstruction error with displacement of the state. A corti-

cal realization must specify the represented variables, their metric and the required fidelity; changing readouts require a further dynamical model.

We propose reflexivity as necessary because even an unreflective episode has a perspective: the cup appears somewhere relative to the perceiver, whether or not the perceiver attends to themselves. Requiring internal self-encoding on that basis is the philosophical hypothesis. Gluing alone does not require it; the dissociations below test the two commitments separately.

5.3 Which cover and which self-model?

Both uniqueness results are conditional on their objects. Gluing gives one global extension of specified compatible local data; Banach gives one fixed point of a specified contracting map. These solve ambiguity within a chosen construction. They do not choose between different covers, different encodings or different candidate subjects. Integrated information theory makes exclusion an explicit part of its account (Tononi et al., 2016); the present uniqueness results do not settle that selection problem.

The relevant cover and map must be tied to neural organization by evidence independent of the desired conclusion. Local descriptions should predict the contents attributed to their domains, and their restrictions should preserve meaning on shared domains. The encoding and readout should be specified before testing them, with their sensitivity to changes in the proposed state measured over a declared range.

A constant readout returning one predetermined state contracts and has a

fixed point, even when it discards an encoding supplied by restriction. Such a construction fails the sensitivity requirement above. This is why selecting the map requires evidence about its functional role in addition to mathematical existence.

5.4 Dissociations that would bear on the identification

A test of the full commitment must measure its clauses separately: effective coupling relative to diffusion, overlap compatibility, and the accuracy of the proposed self-encoding, alongside an independent endpoint for experience. A case meeting all three physical and representational conditions with evidence against a unified conscious episode would challenge their sufficiency. Unified experience with demonstrably incompatible local descriptions would challenge the unity identification. Unified experience with compatibility but without the specified self-encoding would challenge the reflexive requirement.

Anesthetic emergence, dreaming and dissociative states are candidate settings because the framework asks whether coordination, unified content and self-representation can separate. Interpreting such evidence requires a declared relation between the experience endpoint and report; absence of report alone cannot define absence of an episode. The compatibility observable in the supplement is a specification, and the corresponding neural self-encoding has yet to be identified. These are the main obstacles to testing the interpretive commitment.

The proposal concerns which state is conscious. It supplies no explanation

478 of why satisfying these relations should be accompanied by experience. Nor
 479 do its gluing and fixed-point conditions select an electromagnetic substrate:
 480 the cortical field commitment resides in the physical identification and cali-
 481 bration. Their multiple realizability leaves the status of other biological and
 482 artificial implementations open.

483 **6 Thermodynamic constraints and the con-** 484 **ditional composition**

485 The broader composition constrains one proposed physical implementation
 486 through finite-register capacity $H(p) \leq \log |X|$, erasure heat under a specified
 487 system–bath model (Landauer, 1961; Norton, 2005; Shenker, 2000), and a
 488 separate field-boundary result for disconnected vacuum components (Kibble,
 489 1976). The supplement develops these results and their witnesses. The phase
 490 threshold and representational argument stand independently of a derivation
 491 from finite capacity.

492 The constraint most directly relevant to representation concerns predic-
 493 tive memory. For a state X_t driven by an environmental signal S_t ,

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (9)$$

494 Under the no-back-action Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$, the bracket
 495 is nonnegative (Still et al., 2012). The bound limits nonpredictive mem-

496 ory at a given dissipation budget; zero retained information satisfies it too.
 497 Relating it to Landauer heat requires that memory waste be bounded by
 498 erased entropy at the same thermal scale and heat budget. These distinct
 499 information constraints (Rosas et al., 2026) leave content and learning dy-
 500 namics to be specified. Reading either side of such a trade-off from data
 501 is a further problem. In a partially observed signaling model, an agent can
 502 estimate information gathered about a hidden signal and the dissipative cost
 503 of gathering it from the counting statistics of observed trajectories alone, at
 504 finite measurement time and limited resolution (Nicoletti et al., 2026). The
 505 corresponding cortical estimator is not supplied here.

506 An acting agent changes its future inputs, so the passive assumption
 507 requires justification for the variables and interval considered. A feedback
 508 model specifies a policy $A_t \sim \pi(\cdot | X_t)$, world dynamics $S_{t+1} \sim K(\cdot | S_t, A_t)$,
 509 an observation channel and a memory update. For finite states, the exact
 510 information balance is

$$\begin{aligned}
 I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t | S_{t+1}) \\
 &\quad - I(X_t; S_{t+1} | S_t).
 \end{aligned}
 \tag{10}$$

511 Both conditional terms are nonnegative. The last term vanishes for the
 512 passive model; with feedback, the signed difference can be negative because
 513 actions create correlations with the future. Its causal interpretation requires
 514 an environmental state sufficient to account for the relevant dynamics.

515 The finite thermodynamic extension instead bounds joint entropy reduc-

tion. For an elementary update that holds the controller fixed, forward
 paths whose reverses carry positive probability give $\Sigma = D_{KL}(P_F||P_R) =$
 $\Delta H(X, S) + \langle Q \rangle / (k_B T) \geq 0$ under local detailed balance, with Q positive
 into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa, 2013). Strict
 positivity of every mass is sufficient for that condition and is not necessary
 for it, which matters because the channels an agent executes are sparse: a
 step whose initial law sits on a single atom satisfies the support condition
 and fails positivity. A forward path with no reverse, of which deterministic
 erasure is the standard case, has infinite divergence, and the extended-valued
 form records that instead of returning the finite number a totalized logarithm
 supplies. This supplies the thermodynamic constraint for the active branch
 below. A proved two-substep extension composes actuation with a memory
 update initialized from its coordinate-swapped final law. It identifies the
 output with the perception–action channels and cancels the same interme-
 diate entropy and expected energy. At a common thermal scale, the whole
 update’s joint entropy decrease is bounded by its total heat budget divided
 by $k_B T$, without assuming stationarity. The supplement gives the balances
 and reciprocal two-bit witnesses with explicit interaction energy, heat and
 work. Action and observation alias the existing states; additional physical
 registers require further accounting.

An explicit objective also permits finite policy selection. For a supplied
 terminal reward and a finite set of deterministic policies, a nonempty set sat-
 isfying the complete update’s heat budget has a reward maximizer. A two-bit

539 lamp task has a unique feasible optimum that strictly improves on a feasi-
540 ble constant-action baseline; a stronger policy exceeds the budget. Reward,
541 heat and work are evaluated on each policy’s actual composed process, with
542 common initial law, channels, energy and reservoir convention. The base-
543 line meets the heat budget while missing the task target, so the objective
544 supplies content that thermodynamics alone does not. This result compares
545 fixed policies for one update.

546 A separate finite learning model specifies how a policy register adapts. Its
547 bit selects between the two feasible lamp policies, and its energy decreases
548 with their known expected reward. An autonomous thermal channel changes
549 the register, carrying each final distribution into the next update. From the
550 specified initial preparation, Lean proves strictly increasing expected task
551 reward and geometric convergence to a stationary mixture. Transition heat
552 obeys local detailed balance, and cumulative update heat equals the regis-
553 ter’s finite energy loss at zero external work. The mixture remains below the
554 optimal policy; individual updates can lower reward, and a different initial
555 preparation lowers expected reward. This is adaptation toward an encoded
556 objective. Preparing the register and implementing its readout in an actua-
557 tor require additional accounting. Task reward and task heat are prospective
558 expectations under the prepared lamp law; the learning clock executes no
559 task episodes and supplies no spatial coupling convergence or cortical iden-
560 tification.

561 A second learning model supplies no values at all. Which of two actions

562 is rewarded is an environmental bit that no channel of the learner takes as an
 563 argument: the register's update is a function of the observed outcome alone,
 564 and its readout, which fixes the action executed, is a function of the register
 565 alone. The world returns success with probability $3/4$ for the rewarding
 566 action and $1/4$ for the other; success leaves the register where it is and failure
 567 redraws it. Acting, observing and updating compose into one channel on the
 568 joint law of parameter and register, so reward, heat and entropy are read from
 569 the same process. From a uniform prior over both bits, Lean gives every mass
 570 of that law at every horizon: each agreeing pair carries $3/8 - \frac{1}{8}(1/2)^n$, so the
 571 expected reward of the action actually executed is $3/4 - \frac{1}{4}(1/2)^n$, strictly
 572 increasing from $1/2$ and always short of certainty. A frozen register and an
 573 uninformative world, in which success has probability $1/2$ whatever is done,
 574 both hold $1/2$ at zero cost, so the improvement is bought by the observations
 575 rather than by the update rule. Cumulative heat is $(\log 3/4)[1 - (1/2)^N]$, and
 576 the register's two states are energetically degenerate, so that heat is external
 577 work. It is irreducibly so: a register energy that made the updates free would
 578 force the drive's heat to be equal at both parameter values, and this drive's is
 579 not. The learner acquires no preference between the actions, each of which it
 580 executes half the time at every horizon, but a correlation with the parameter.
 581 Preparing the prior, supplying the work, executing further task episodes and
 582 fabricating the readout are separate accounting, and no cortical identification
 583 follows.

584 6.1 What the composition establishes

585 A continuing finite agent sequences actuation, observation-driven learning
586 and reset on one joint law of an unknown task parameter, a register and a
587 task flag. The register drives the actuator; the observation reads the flag
588 that action changed; the reset prepares its next use. Each stage starts from
589 the preceding output. For a supplied initial work store B , a common energy
590 E and thermal scale θ , the first law and entropy balance telescope to

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (11)$$

591 provided expected cumulative work stays within B at every prefix and each
592 stage satisfies positive support and local detailed balance. A bit witness
593 retains agreement with the unknown rewarding action above chance after
594 every completed cycle, while every cycle draws a positive minimum of work,
595 including reset. Blinding the observations leaves chance performance and
596 makes the learning stage free. Replacing reset by idle drift preserves some
597 learning but reduces the next cycle's performance. A finite store therefore
598 supports only a bounded prefix of this protocol. This is accounting in expec-
599 tation: the initial law and store are supplied resources, and the observation
600 is marginalized inside the learning channel. Pathwise stores, a separate sen-
601 sor memory and preparation channels are treated below. Heat is assigned to
602 the composite learning channel through local detailed balance, and the store
603 bounds a mean; no cortical identification follows.

604 A store the protocol carries rather than reports supports a different claim.
 605 Let the state include the store's reading β , and let each executed transition
 606 draw w^- from it and receive a declared supply w^+ , with

$$\beta(t) = \beta(s) - w^-(s, t) + w^+(s, t), \quad w^-(s, t) \leq \beta(s) + w^+(s, t) \quad (12)$$

607 on every transition of positive probability out of a state the protocol reaches.
 608 A reading that starts nonnegative and satisfies the second condition at every
 609 stage is nonnegative at every state the protocol reaches, at every horizon.
 610 Averaging the first identity over the executed paths returns the mean ledger,
 611 $\langle \beta \rangle_N = \langle \beta \rangle_0 - W_{<N}^- + W_{<N}^+$, so pathwise solvency implies the expected bound
 612 Eq. (11) assumes. The converse fails: a store that spends a unit or idles with
 613 equal probability covers its expected draw at both prefixes of a two-stage
 614 run, with equality at the second, and is overdrawn on the path of probability
 615 $1/4$ that spends twice. An expected allowance is a bound on a mean and not
 616 a battery. Two conditions fence sustained operation. If the supply covers
 617 each draw as it is made, every horizon is solvent and no bound on cumulative
 618 work is needed. If instead $w^- - w^+ \geq c > 0$ on every reachable transition,
 619 a reading starting no higher than β_0 gives $Nc \leq \beta_0$, which is the pathwise
 620 form of the store bound and constrains each trajectory rather than a mean.
 621 A deterministic spend-and-recharge cycle satisfies neither condition and is
 622 solvent at every horizon while the work it has drawn exceeds any declared
 623 allowance, so continuing operation is a claim about the supply. None of this

624 requires positive support; the generic store accepts its supply as an input.

625 Neither the ledger nor the horizon it bounds needs a finite state space. On
626 any state type a reading obeying the first identity of Eq. (12) telescopes to
627 $\beta_N = \beta_0 - W_{<N}^- + W_{<N}^+$, so a nonnegative final reading bounds the cumulative
628 draw, and a uniform net cost c gives $Nc \leq \beta_0$; the expected-work horizon
629 above is that bound read on the sequence of allowances rather than a second
630 argument. What the finite state space is for is the pathwise statement,
631 which quantifies over reachable states instead of following one sequence. Two
632 instances use the room. A real-valued reading whose delivery matches its
633 draw is solvent at every operation, which is what refuelling from outside looks
634 like while the supply is declared rather than derived. And on a countably
635 infinite state space, swapping a source's charge into a buffer is an involution
636 conserving the total, so transferring n units of charge is reversible and costs
637 nothing; preparing that charge is a separate problem, on which this bears
638 not at all.

639 A finite source can be included in the same state. If its usable energy R
640 pays for each supplied transfer, then on every reachable transition

$$R' = R - w^+, \quad (\beta + R)' = (\beta + R) - w^-. \quad (13)$$

641 Internal replenishment cancels from the combined ledger. Its expected supply
642 is $W_{<N}^+ = \langle R \rangle_0 - \langle R \rangle_N$, and nonnegative store and source readings give
643 $W_{<N}^- \leq \langle \beta + R \rangle_0$. If every executed draw before horizon N is at least $c >$

644 0 and the initial combined reading is at most b , then $Nc \leq b$. A direct
 645 transfer of one unit followed by idling saturates this bound; the cost premise
 646 concerns the finite run. Available energy does not guarantee delivery: a
 647 finite witness starts a source bit charged and its buffer and output load
 648 empty, with a uniform selector bit. The selector controls a source–buffer
 649 swap, followed by a buffer–load swap. Both gates are reversible and conserve
 650 the three energy bits’ total energy. Half the paths deliver a unit to the load;
 651 half leave that energy in the source. The actual mean draw is $1/2$, and
 652 claiming a unit on every path violates the store ledger. An initially empty
 653 source delivers nothing. The indefinitely replenished charger cannot have
 654 any nonnegative finite source coordinate satisfying Eq. (13). This witness
 655 models one uncertain transfer followed by idling; source preparation, gate-
 656 control costs and identification with the learning agent’s actuator remain
 657 separate, and the external delivery of the trajectory ledger is declared rather
 658 than sourced.

659 The measurement has a register of its own, and clearing it is an executed
 660 operation. Let the state carry a memory coordinate alongside the register and
 661 the world, and run four operations in sequence: the readout drives the actua-
 662 tor, a measurement writes the memory from the world the action moved, the
 663 register is updated from the memory alone, and a clearing operation drives
 664 the memory toward a declared standard law ν while the world is prepared.
 665 The update takes the memory and not the world, so two worlds holding the
 666 same memory drive the register identically: what the register can learn is

667 what the memory carries. The clearing operation drives three coordinates at
 668 once, and its transition log-ratio is the sum of the three coordinates' own, so
 669 the memory's share is a named part of that operation's cost rather than a
 670 residue. For an erasure that lands on ν whatever it is given, that share is

$$\langle Q_M \rangle = \theta[H(\mu) - H(\nu)] + \theta D_{KL}(\mu||\nu), \quad (14)$$

671 with μ the memory's marginal before the operation. The first term is Lan-
 672 dauer's bound and the second is the cost of applying one fixed channel to a
 673 law it is not matched to; Gibbs' inequality turns the identity into the bound.
 674 The identity also states what the bound does not. A memory already at ν
 675 is free to clear, and one more ordered than ν draws heat out of the reser-
 676 voir, so clearing dissipates because of what the memory holds rather than by
 677 construction.

678 A bit witness fixes the numbers without computing the joint law. The
 679 memory's marginal after a measurement is a mixture of measurement outputs
 680 over a world marginal that is itself a mixture of actuator outputs; with every
 681 channel mass $1/4$ or $3/4$, the memory's mass on a set flag lies in $[7/16, 9/16]$ at
 682 every clearing, whatever the agent has learned. At $\nu = (3/4, 1/4)$ the share is
 683 $[3/4 - \mu(0)] \log 3$, hence between $(3/16) \log 3$ and $(5/16) \log 3$ at every cycle:
 684 no amount of learning makes the measurement free. In the first cycle the
 685 memory is exactly uniform and the two terms of Eq. (14) are $(3/4) \log 3 - \log 2$
 686 and $(1/2) \log(4/3)$, summing to $(1/4) \log 3$; the first is positive because $16 <$

687 27. Two controls fence the reading. A blind measurement that records
 688 nothing about the world leaves the same marginal and pays the identical heat,
 689 so the cost is set by the memory's marginal and not by what the memory
 690 is about. Replacing the clearing by the symmetric drift costs nothing and
 691 carries the measurement into the next episode, where the erasure returns the
 692 standard state whatever was written. The four operations are stages of one
 693 protocol, so Eq. (11) charges the whole run, clearing included. Fabricating
 694 the gates and any cortical identification remain outside this model; preparing
 695 the prior is charged below by the same identity.

696 The work those operations draw can be carried in the same state. Let a
 697 coordinate R of the agent's own state read a finite charge, let each executed
 698 transition draw that coordinate's loss, and let the clearing spend at least c of
 699 it while no operation returns any. If the charge starts no higher than b and
 700 stays nonnegative through m complete cycles, then

$$mc \leq b, \tag{15}$$

701 so a finite source funds finitely many clearings. The bound needs a cost that
 702 depends on the stage: an agent that pays at one operation in four meets
 703 a uniform per-transition cost only at $c \leq 0$, while a charge paid once in
 704 each period of p operations sums to mc over m periods however it sits inside
 705 the period. A witness whose world carries the task flag and a charge of
 706 three units, one spent at each clearing, attains Eq. (15) at three cycles and

707 reads below empty at the sixteenth operation. Its clearings cost between
708 $(3/16)\log 3$ and $(5/16)\log 3$ throughout, by the marginal bounds above, so
709 what the source funds is what Eq. (14) prices. The drawn unit is a declared
710 reading of the state and is not identified with that heat.

711 Two limits here belong to the model rather than to the witness. The
712 clearing is a channel of the memory alone, so it cannot be conditioned on
713 the charge: past the third cycle the agent still clears, still costs at least
714 $(3/16)\log 3$ and draws nothing, so the run is unfunded rather than halted.
715 And an agent all of whose channels have full support cannot be funded at
716 all. A ledgered reading that never rises is then constant between every pair
717 of states, in both directions, so every net draw is zero, and the witness's
718 restricted support is forced by that rather than chosen. Preparing the charge
719 and fabricating the gates remain outside this model, and the declared delivery
720 above is not connected to this agent's channels.

721 Preparation can be modelled by the same constant channel as memory
722 clearing. For a strictly positive target ν , Eq. (14) gives its mean log-ratio
723 heat from hardware law μ as $\theta[H(\mu) - H(\nu)] + \theta D_{KL}(\mu\|\nu)$. Against a uni-
724 form target the two terms cancel, because $D_{KL}(\mu\|\text{unif}) = \log |S| - H(\mu)$:
725 the channel has zero mean heat from any starting law. The reservoir inter-
726 pretation uses local detailed balance; gate construction, control work and a
727 funding source require separate accounting. Prepending this channel as stage
728 zero includes preparation in the cumulative balance whenever its hypotheses
729 hold. Once it reaches the declared prior, every subsequent law is the original

730 agent’s own. The sensor-memory witness prepares its uniform prior from a
 731 different, strictly positive hardware law and satisfies the whole-run entropy
 732 bound.

733 Two things a preparation cannot do are properties of its signature. It
 734 writes the system coordinate and leaves the parameter’s marginal as it found
 735 it, so the agent’s uncertainty about the world is an input rather than some-
 736 thing it prepares; and a preparation that reads neither coordinate produces
 737 a product, so it cannot create a correlation between parameter and regis-
 738 ter. At $\theta = 1$, preparing a bit law biased to ε from uniform has mean heat
 739 $(1/2 - \varepsilon) \log[(1 - \varepsilon)/\varepsilon]$ for $0 < \varepsilon < 1$; this family has no finite upper bound. Its
 740 sharp endpoint has zero reverse probabilities and fails the identity’s positive-
 741 target premise. This is a restriction of the specified channel and heat model;
 742 it supplies no cost bound for other implementations of exact preparation,
 743 including the charge above.

744 Table 1 lists the connecting hypotheses of two branches. The theorem
 745 **chain** uses passive predictive memory; **chain_active** uses a specified finite
 746 feedback process. Each accepts eight implications and shares the passage
 747 from coarse-graining to the phase model and its proposed representational
 748 role. In both, the finite register and the field boundary are independent
 749 starting objects.

750 For the active branch, the controller has the register’s state type and the

751 fourth node records

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (16)$$

752 where p, p' and $\langle Q \rangle$ belong to that same controlled substep. E34Active sup-
753 plies positive path probabilities, local detailed balance at the register's ther-
754 mal scale θ , and the allocation $\langle Q \rangle \leq B$, with B equal to the register's
755 heat. Landauer's lower bound does not imply this upper-budget allocation
756 or identify erasure with actuation. A resource model of one register supplies
757 it instead: every operation is controlled by that register and exchanges heat
758 with its reservoir at θ , and the register's dissipated heat is the total its op-
759 erations deliver there. An operation that does not increase the joint entropy
760 then has a nonnegative share, by the left inequality of Eq. (16) applied to it,
761 so an operation whose companions all compress has a share no larger than
762 the register's dissipation. That accounting identity and that compressiveness
763 are the model's physical inputs; the allocation follows from them. Composing
764 the operations into one evolving law with its preparation and power source
765 is a separate model, and the register–bath model at the end of this sub-
766 section shows that a microscopic bipartite dynamics supplies neither input.
767 E45Active requires convergence of the specified coupling energies. One scalar
768 constitutive model connects those energies to the named feedback step: writ-
769 ing $d = H(p) - H(p')$, it sets the spatial density to $f_A(x) = g db(x)$, with

770 supplied gain $g \geq 0$ and profile b . Regular spatial refinement gives

$$E_n(f_A) \longrightarrow L_A = g d \int_S b d\mu, \quad L_A \leq g \frac{B}{\theta} \int_S b d\mu \quad (17)$$

771 when $\int_S b d\mu \geq 0$. The limit follows from mesh assumptions; the upper
 772 bound uses Eq. (16) for the same process. The constitutive law uses ensemble
 773 entropy reduction and supplies neither a microscopic actuator nor its work
 774 account. Changing the gain or profile at fixed process and heat changes the
 775 coupling energies.

776 A second, microscopic model supplies an actuator and a kernel. The
 777 hardware carries finitely many spatial response modes with fixed profiles ϕ_i
 778 and installation prices u_i . A configuration z of the controller and its switches
 779 sets the mode occupancies $c_i(z)$, and those same occupancies determine both
 780 a reciprocal kernel on the product space and a stored installation energy,

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (18)$$

781 A transition $z \rightarrow z'$ changes K only through the modes whose occupancy it
 782 changes, so a pair outside the response of those modes is untouched, and the
 783 transition's external work is $U(z') - U(z) + Q$. The kernel the executed step
 784 installs is the average of K_z over the law that step actually leaves; its change is
 785 the mean of the pathwise changes, and the mean of the transition works splits
 786 into the change in stored energy and the channel's heat. A finite measurable
 787 partition of the substrate with one sample per cell reconstructs the installed

788 kernel throughout each cell pair, and the cell-mass weighted finite energy
789 equals the product-measure integral of that reconstruction exactly. On a
790 compact substrate with continuous profiles, a sample error tending to zero
791 gives

$$\mathcal{E}_n \longrightarrow \iint K d\mu d\mu. \quad (19)$$

792 The refinement index here is spatial and is independent of the number of ex-
793 ecuted control updates; convergence in one is not convergence in the other.
794 The same hardware discharges the passive edge from a declared predictive
795 system's own joint law, and two declared systems with different dissipated
796 work give the identical edge, so predictive efficiency selects neither the hard-
797 ware nor the partition. Mode profiles, prices, occupancy readout, reservoir
798 convention and substrate are declared hardware data, which no heat bound
799 supplies. Fabricating that hardware, powering it beyond one prepared up-
800 date, and identifying the substrate or the modes with a physical field remain
801 separate requirements.

802 A third finite model examines the register's own bath. One register bit
803 and one bath bit carry each of the four deterministic maps of the register as
804 a permutation of the pair, acting on a bath prepared in its ground state: an
805 injective map moves the register alone, and an erasure swaps the register's
806 bit into the bath. With bath levels 0 and $2 \log 2$, degenerate register levels, a
807 uniform register and $\theta = 1$, the bath's mean energy gain along the executed
808 paths is 0 for each injective map and $\log 2$ for each erasure, and those values

809 inhabit Landauer's heat equation with the bath supports the gates actually
810 reach. For an arbitrary finite protocol and an explicitly modelled bath energy
811 E_B ,

$$\sum_{k < N} \langle \Delta E_B \rangle_k = \langle E_B \rangle_N - \langle E_B \rangle_0, \quad \sum_{k < N} \langle q_k \rangle = \langle E_B \rangle_N - \langle E_B \rangle_0 + \sum_{k < N} \langle q_k - \Delta E_B \rangle, \quad (20)$$

812 where the first identity needs no positive support and supplies no entropy
813 inequality, and the second is the correction a reported heat ledger carries
814 when it omits part of that transfer. Three consequences fence what a map-
815 indexed budget can mean. Executing the same erasing gate twice returns
816 the $\log 2$ and restores the initial law, so the two shares are $\log 2$ and $-\log 2$
817 about a total of zero, and a reusable eraser requires its bath prepared again. A
818 second reversible gate holds the register fixed and excites its bath, performing
819 the same register map as idling at $2 \log 2$ instead of 0; the work supplied is
820 $\log 2$ for one erasure, $2 \log 2$ for that drive and zero for two erasures. And a
821 swap is an involution, so the gates' own transition log-ratios vanish on every
822 realized path while their bath gains $\log 2$: a deterministic reversible gate is
823 not a strictly positive reduced channel, and local detailed balance at any
824 thermal scale assigns it none of the heat its bath receives. Positivity, local
825 detailed balance and the ledger identity are inputs of the resource model
826 rather than consequences of microscopic reversibility. Preparing the bath,
827 supplying the work, and identifying this bath with the reservoir of the ledger's
828 operations remain separate; the numerical values are those of the declared

829 energy gap and preparation.

830 Two structures also enter both theorems: a thermodynamic cover, in-
831 cluding overlap compatibility, and a reflexive boundary specifying the local
832 encoding and readout. Given these and the connecting hypotheses, both
833 conclude **UnifiedSelf**: the section glued on that cover is the unique fixed
834 point of self-prediction. Restriction resonance, needed for the interpretation
835 in Eq. (7), is a separate condition. The result binds the fixed point to the
836 specified local states; compatibility and accuracy at that state are premises,
837 while uniqueness follows from the construction.

838 The joint witnesses combine a double well, a register, a changing uniform-
839 grid mesh and a three-site cortex; the active branch includes the noisy two-bit
840 actuator. They establish that the hypotheses can hold together in distinct toy
841 components. Mesh convergence is supplied independently of the thermody-
842 namic bounds, and no common biological mechanism is established. A second
843 active witness composes the same branch with the bridge discharged from
844 that register's ledger: its two operations deliver $(2/5) \log 2$ and $(3/5) \log 2$ to
845 one reservoir, summing to the register's $\log 2$. A third active witness closes
846 one spatial response mode with a thermal switch: executing the channel
847 strictly raises a pairwise kernel value, closing the mode costs its stored en-
848 ergy plus the channel's heat, and uniform partitions of an atomless interval
849 give a coupling energy that changes between the first two refinements and
850 converges to the value the field witness uses. The active witnesses reject
851 a zero heat budget, an incompatible energy limit, a compressive operation

Connection	Status	Required assumption
E12	Independent physical premise	A finite-capacity system realizes a field that leaves its vacuum manifold. Capacity and the boundary are independent roots.
E23	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active	Bridge assumption	Passive: a predictive structure at the register’s thermal scale and heat budget, with nonzero wasted memory. Active: positivity, local detailed balance and a register-heat allocation, derived from its ledger identity and other operations’ compressiveness. Preparation, source and store accounts use their own laws; identifying them with register heat remains an input. A reversible register–bath gate realizes erasure without satisfying the positivity or local-detailed-balance premise.
E45 / E45Active	Modelling assumption	Passive: a convergent regime, for hardware holding a declared predictive law. Active: Eq. (17) from a supplied gain, profile and meshes; Eq. (19) from declared response modes.
E56	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67	Physical commitment	That field operates above the synchronization threshold.
E78	Bridge assumption	Coherence drives a supplied cover to its relaxed equilibrium at coupling at least the named mean-field value.
E89	Modelling assumption	A map mediated by local self-encoding fixes the section glued on that cover and has the Lipschitz factor $q < 1$ used in the exact and approximate-state results.

Table 1: Connecting hypotheses of the passive and active compositions, eight per branch. They share E12, E23 and E56–E89. The cover’s compatibility is supplied as a structure field. Identifying the boundary’s encoding with restriction requires a separate resonance condition. The mathematical objects and node-level results are mapped in Supplemental Material, Table S1.

852 above the register’s whole dissipation, a ledger whose other operation draws
853 heat out of the reservoir, an uncharged installation, and a wrong spatial limit
854 for the microscopic sequence. Counterexamples also expose the role of over-
855 lap agreement. The development declares no additional Lean axioms beyond
856 the foundational axioms used by its libraries; physical premises are explicit
857 hypotheses and model fields.

858 **7 Discussion**

859 The strongest results concern what a coherence measurement leaves unre-
860 solved. Different recovery mechanisms share stationary distributions; equal
861 phases permit incompatible local descriptions; and the tested plasticity ob-
862 jective sustains coherence while moving away from its diagnostic structure.
863 These findings make the coupling hypothesis more demanding to test and
864 prevent a successful synchronization model from doing unexplained repre-
865 sentational work.

866 The proposed account identifies a compatible, internally represented state
867 with a conscious episode. Its main explanatory gap is the mechanism that
868 would maintain those relations: compatibility and accuracy at the glued state
869 enter the construction as premises. Their mathematical consequences leave
870 open why cortical activity should acquire or preserve them. The two halves
871 of that question separate. Carrying patch contents as fields and mixing each
872 with its own observation at gain η answers the second: observations that

873 agree across an overlap preserve exact agreement and contract any mismatch
 874 to $(1 - \eta)^n$ of itself after n of them, while observations differing by δ leave
 875 a residual floor at δ rather than none. The first half is answered by the
 876 coherence the rest of the account is about. The order parameter satisfies
 877 $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$, and because every term of $1 - \cos$ is nonnegative
 878 a single pair obeys $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$: a population average
 879 bounds each individual phase gap, at a factor N^2 that is the price of a point-
 880 wise statement drawn from a mean. Contents that are a declared Lipschitz
 881 function of the local phase, read as a point of the circle rather than as a
 882 real number, are then compatible on every overlap at a residual computed
 883 from r alone, with no compatibility premise anywhere and exact agreement
 884 at $r^2 = 1$. What the derivation supplies is agreement; what it still declares
 885 is the encoder, and the contents remain real-valued fields identified with no
 886 neural variable.

887 The most informative next step for that connection is a test of whether
 888 coordination helps maintain independently measured content agreement. Af-
 889 ter validating separate decoders and their overlap restrictions, a controlled
 890 change in coupling or coordination could test whether agreement deterio-
 891 rates in the predicted way, with controls for changes in sensory information
 892 and decoder quality. Such a result would bear directly on the proposed role
 893 of coherence. In parallel, the awakening hypothesis needs its own calibrated
 894 geometry–coupling relation and resolved phase dynamics to distinguish a cou-
 895 pling increase from diffusion change or common drive. A favorable recovery

896 trace alone would leave the representational connection open.

897 Testing the self clause requires perturbing a candidate encoding and com-
898 paring the resulting changes in represented scene, body and viewpoint with
899 the readout’s predictions. Establishing that functional role would make dis-
900 sociations from unified experience informative. The exact fixed-state ideal-
901 ization must also be assessed against the fidelity and time dependence of the
902 biological process.

903 Five components of the account are not derived here, and the formaliza-
904 tion marks each. Coarse-graining the discrete couplings converges a scalar
905 energy and produces no kernel: the vertex-sited candidate contributes ex-
906 actly zero continuum coupling energy on an atomless substrate, which is a
907 no-go for that architecture rather than for kernels, since cell pairs of pos-
908 itive mass do pass to a kernel limit. The field conversion $K_{\text{eff}} = \gamma N s E f$
909 carries its physical content in γ , a hypothesis about how long field-induced
910 timing shifts persist; the oscillation frequency, a phase relaxation rate and
911 the diffusion constant are candidates that need not agree, and the remaining
912 factors cannot choose between them. Descent of the squared-drift objec-
913 tive is proved for the specified flow at fixed phases, whereas the claim that
914 such adaptation acquires a designated environmental structure is not proved
915 and fails in the bounded study reported above. The non-identifiability re-
916 sult concerns stationary distributions, so it limits what a coherence trace
917 establishes rather than what any measurement can establish; separating the
918 mechanisms requires resolving dynamics. The thermodynamic results take a

919 prepared initial law as given and leave the supply of free energy to a sepa-
920 rate model. These are boundaries of scope and derivation. Reading any of
921 them as a positive finding about cortex, or as support for scale separation,
922 maladaptive learning or a limit on observation in general, would overstate
923 what has been checked.

924 Two of these results bear on formulations that read a generative model
925 off a system’s steady state, as the free-energy principle does (Friston, 2010).
926 Nothing here refutes that principle on its own terms: the squared-drift func-
927 tional is not a variational free energy, and no bridge to one is claimed or
928 needed for any theorem above. It is also not thermodynamic entropy pro-
929 duction, and that is now a theorem rather than a caution: at the coherent
930 stationary state the current dissipation over phase space is zero while the
931 phase-averaged squared drift there is positive, so the two functionals differ
932 on the same argument, and the substrate-level objective of the plasticity
933 results is a functional of a third. What transfers is narrower. The station-
934 ary distributions of increasing coupling, decreasing diffusion and a matched
935 common drive coincide, so a steady-state density does not by itself deter-
936 mine which mechanism produced it, and here the competing mechanisms are
937 written down rather than supposed. Separately, local descriptions agreeing
938 pairwise on their overlaps need not determine one global state, and at ev-
939 ery positive tolerance a three-patch family admits no global section at all;
940 a decomposition into local descriptions therefore does not by itself deliver
941 one belief about one world. Both points instantiate identifiability concerns

942 already raised about that principle (Aguilera et al., 2022; Biehl et al., 2021;
943 Bruineberg et al., 2022), in a model where the alternatives are explicit. Nei-
944 ther establishes that cortex fails to perform inference, and the plasticity
945 control above bears on the objective it names and on no other.

946 Integrated-information, global-workspace and predictive-processing ac-
947 counts organize unity, access and modelling differently (Tononi et al., 2016;
948 Baars, 1997; Dehaene et al., 2014; Friston, 2010). The present proposal
949 contributes a candidate interaction mechanism and explicit relations to test
950 between local content and internal self-encoding. Its explanatory value will
951 turn on whether those relations can be identified independently and shown
952 to matter for experience. For the cup example, that means explaining how
953 the system maintains one represented situation and how the perceiver’s place
954 within it contributes to the episode.

955 **Data and code availability**

956 The Lean development, simulation scripts, tests and saved numerical sum-
957 maries are maintained in the manuscript source repository. Publication sim-
958 ulation values are regenerated from saved summaries without rerunning pro-
959 duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset
960 ds005620 (Bajwa et al., 2025); preprocessing and selection are described
961 in Supplemental Material. The source repository contains the reproducible
962 analysis and does not redistribute the raw recordings.

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966 **Use of AI tools**

967 AI assistance to this work was substantial and is not separable by task.
968 Several coding assistants and chat interfaces were used in an interleaved way
969 throughout: Google’s Antigravity CLI (Gemini 3.1, Claude 4.6 Opus, Claude
970 4.6 Sonnet), OpenAI’s Codex CLI (Astra, Terra, Luna, Sol), Claude Code
971 (Claude Opus 5), Gemini Pro through its web interface, and Fable (Claude
972 Opus 4) through OpenRouter, accessed with the Hermes CLI. Between them
973 they contributed to the Lean development, the simulation and analysis code,
974 and the drafting and revision of both documents. Because the tools were
975 used in combination and in alternation, no section, proof, figure or result is
976 attributable to any one of them, and none is claimed to be. The author is
977 responsible for all content, including everything a tool produced. Machine
978 checking establishes the stated Lean results under their hypotheses; it does
979 not validate biological interpretations or replace scientific review of generated
980 code and prose.

References

- Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable electric fields emerge from and allow representational drift. *NeuroImage*, 253, 119058.
- Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7), 450-461.
- Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the Mind. *Oxford University Press*.
- Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a computational theory of conscious processing. *Current Opinion in Neurobiology*, 25, 76-84.
- Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature Reviews Neuroscience*, 11(2), 127-138.
- Aguilera, M., Millidge, B., Tschantz, A., & Buckley, C. L. (2022). How particular is the physics of the free energy principle?. *Physics of Life Reviews*, 40, 24-50.
- Biehl, M., Pollock, F. A., & Kanai, R. (2021). A technical critique of some parts of the free energy principle. *Entropy*, 23(3), 293.

1000 Bruineberg, J., Dolega, K., Dewhurst, J., & Baltieri, M. (2022). The Em-
1001 peror’s New Markov Blankets. *Behavioral and Brain Sciences*, 45, e183.

1002 Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
1003 memory network formation. *Cerebral Cortex*, 33(17), 9877-9890.

1004 Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:
1005 Electric fields sculpt neural activity and ”tune” the brain’s infrastructure.
1006 *Progress in Neurobiology*, 226, 102465.

1007 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
1008 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
1009 9(8), 45-60.

1010 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
1011 testable hypothesis about the characteristics of conscious as opposed to
1012 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.

1013 McFadden, J. (2020). Integrating information in the brain’s EM field: the
1014 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
1015 niaa016.

1016 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
1017 guide neocortical network activity. *Neuron*, 67(1), 129-143.

1018 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
1019 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
1020 95-103.

- 1021 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
1022 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.
- 1023 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
1024 on brain circuits in rats and humans. *Nature Communications*, 9, 483.
- 1025 Verkhratsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
1026 *logical Reviews*, 98(1), 239-389.
- 1027 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
1028 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
1029 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.
- 1030 Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J.
1031 F., & Juel, B. E. (2025). A repeated awakening study exploring the capac-
1032 ity of complexity measures to capture dreaming during propofol sedation.
1033 *Scientific Reports*, 15, 32012.
- 1034 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... &
1035 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
1036 brain. *Science*, 342(6156), 373-377.
- 1037 Syková, E., & Nicholson, C. (2008). Diffusion in brain extracellular space.
1038 *Physiological Reviews*, 88(4), 1277-1340.
- 1039 Barbour, B. (2017). Analysis of claims that the brain extracellular impedance
1040 is high and non-resistive. *Biophysical Journal*, 113(7), 1636-1638.

1041 Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return
1042 of Maxwell's demon. *Studies in History and Philosophy of Science Part B:*
1043 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.

1044 Landauer, R. (1961). Irreversibility and heat generation in the computing
1045 process. *IBM Journal of Research and Development*, 5(3), 183-191.

1046 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
1047 *of Physics A: Mathematical and General*, 9(8), 1387-1398.

1048 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
1049 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.

1050 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
1051 of prediction. *Physical Review Letters*, 109(12), 120604.

1052 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
1053 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.

1054 Horowitz, J. M., & Esposito, M. (2014). Thermodynamics with continuous
1055 information flow. *Physical Review X*, 4, 031015. doi:10.1103/PhysRevX.4.
1056 031015.

1057 Ito, S., & Sagawa, T. (2013). Information thermodynamics on
1058 causal networks. *Physical Review Letters*, 111, 180603. doi:10.1103/
1059 PhysRevLett.111.180603.

- 1060 Corberi, F., dello Russo, S., Messuti, G., Scarpetta, S., & Smaldone, L.
1061 (2026). Thermodynamic learning. *arXiv preprint*. arXiv:2609.04732.
- 1062 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
1063 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
1064 181.
- 1065 Shenker, O. R. (2000). Logic and entropy. *PhilSci-Archive preprint*. philsci-
1066 archive.pitt.edu/115.
- 1067 Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Boyd, A. B., Berta, M., Jensen,
1068 H. J., Gastpar, M., & Carroll, S. M. (2026). Four faces of information in
1069 natural systems. *Nature Reviews Physics*. doi:10.1038/s42254-026-00971-4.
- 1070 Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de*
1071 *Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics,
1072 vol. 1464, pp. 165–251. Springer. doi:10.1007/BFb0085169.
- 1073 Berglund, N., & Gentz, B. (2002). Pathwise description of dynamic pitchfork
1074 bifurcations with additive noise. *Probability Theory and Related Fields*,
1075 122, 341–388. doi:10.1007/s004400100174.
- 1076 Abramsky, S., & Brandenburger, A. (2011). The sheaf-theoretic structure
1077 of non-locality and contextuality. *New Journal of Physics*, 13, 113036.
1078 doi:10.1088/1367-2630/13/11/113036.
- 1079 Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin,
1080 P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave

- 1081 patterns in primate cerebral cortex. *The Journal of Neuroscience*, 35(11),
1082 4657-4662. doi:10.1523/JNEUROSCI.4509-14.2015.
- 1083 Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave pat-
1084 terns underlie complex brain dynamics and are related to cognitive pro-
1085 cessing. *Nature Human Behaviour*, 7(7), 1196-1215. doi:10.1038/s41562-
1086 023-01626-5.
- 1087 Nicoletti, G., Di Terlizzi, I., & Busiello, D. M. (2026). Balancing information
1088 and dissipation with partially observed fluctuating signals. *Physical Review*
1089 *Letters*, 137(12), 127101. doi:10.1103/8gx2-rbns.
- 1090 Kiyooka, D., Oomoto, I., Kitazono, J., Saito, Y., Kobayashi, M., Matsub-
1091 ara, C., Kobayashi, K., Murayama, M., & Oizumi, M. (2026). Single-
1092 cell resolution functional networks during unconsciousness are segre-
1093 gated into spatially intermixed modules. *Cell Reports*, 45(2), 116902.
1094 doi:10.1016/j.celrep.2025.116902.
- 1095 Oomoto, I., Kiyooka, D., Oizumi, M., & Murayama, M. (2026). Multi-area
1096 single-cell calcium imaging dataset of the mouse cortex across wakefulness,
1097 sleep, and anesthesia. *Scientific Data*. doi:10.1038/s41597-026-08202-2.